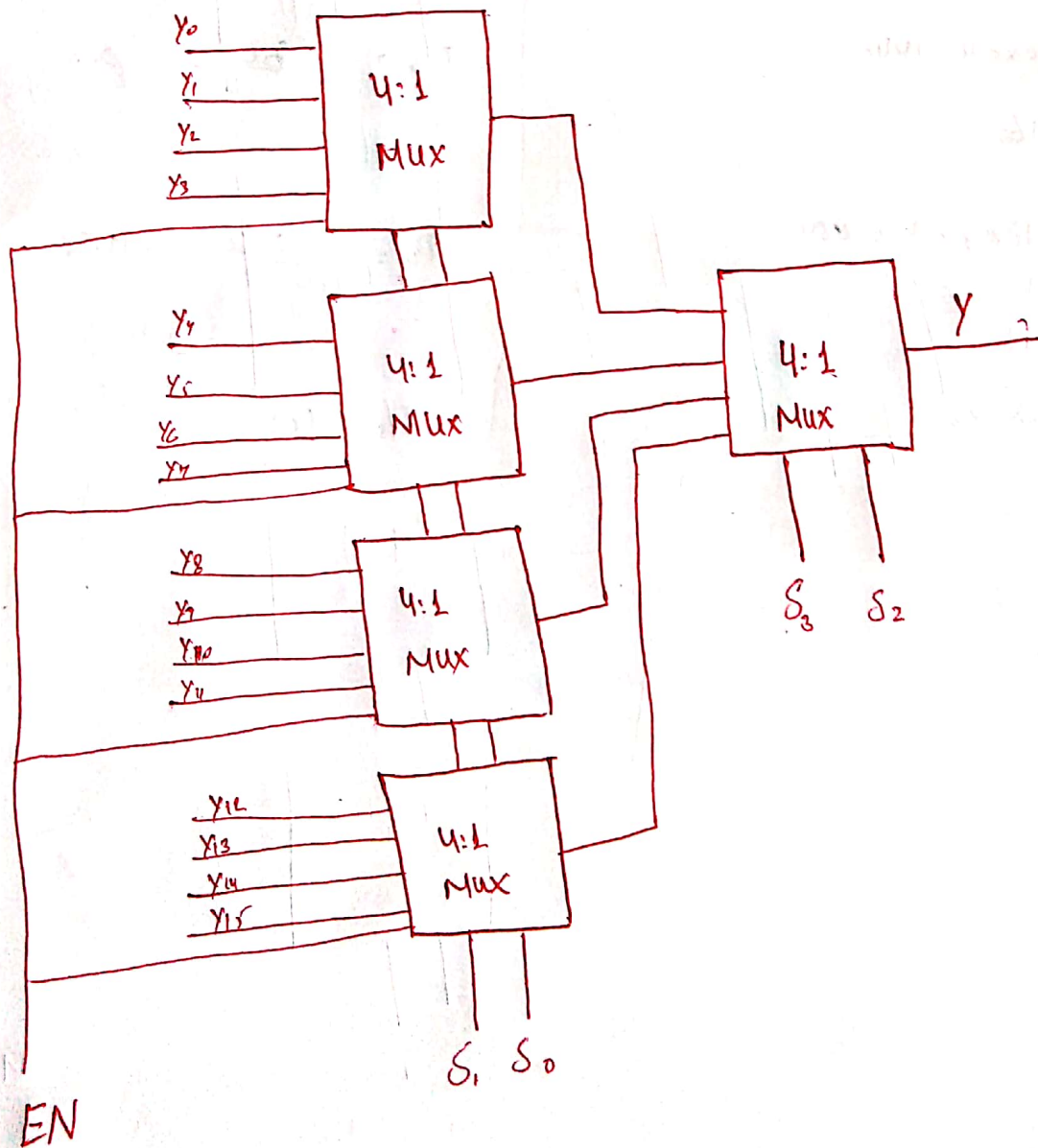
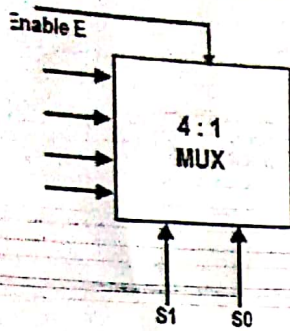


Name: _____

Roll No: _____

Section: _____

Use following 4 X 1 multiplexers and OR gate to create 16 X 1 multiplexer?



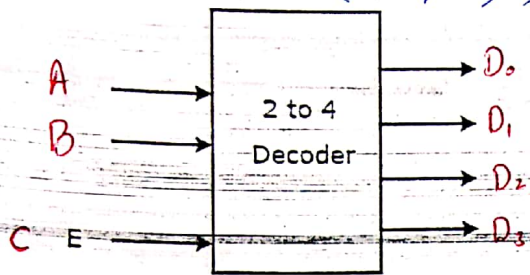
Name: _____

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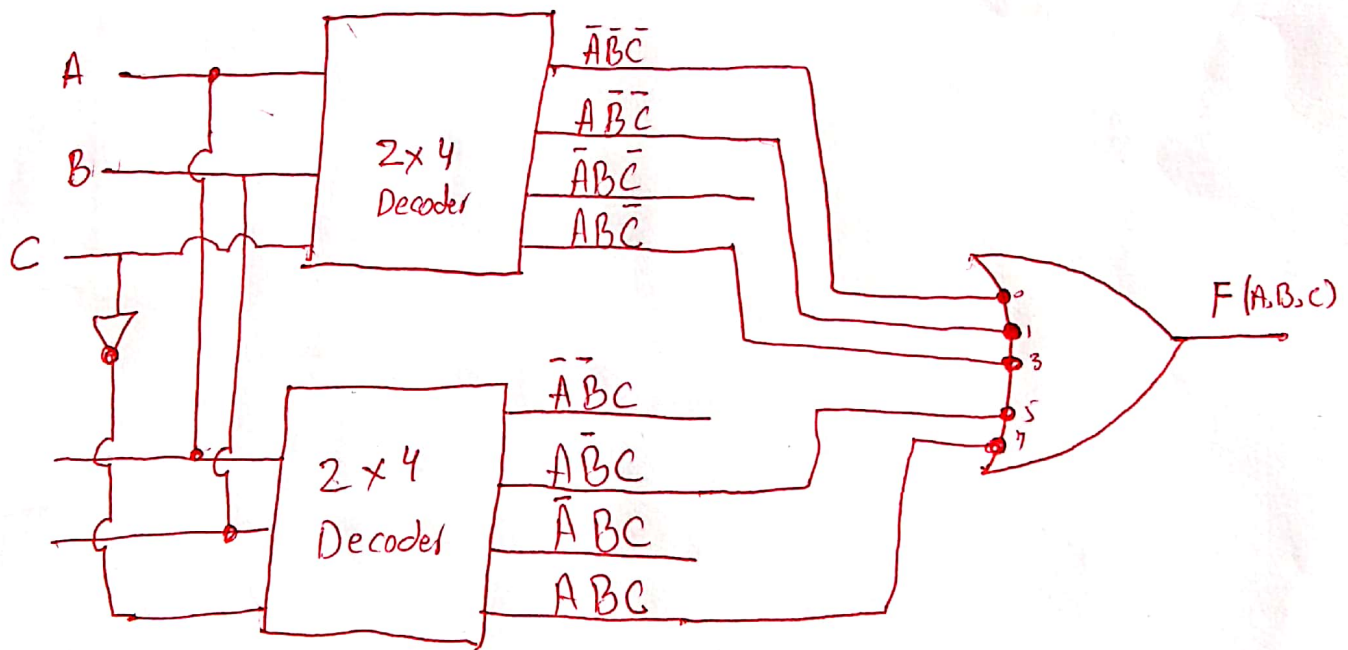
Section: _____

Implement the following function using 2x4 decoders.

$$F(A,B,C) = \sum(0, 1, 3, 5, 7)$$



A	B	C	F
0	0	0	1
1	0	0	1
0	1	0	0
1	1	0	1
0	0	1	0
1	0	1	1
0	1	1	0
1	1	1	1



Name: _____

Roll No: _____

Section: _____

Design 4x2 Random Priority Encoder with following input priority: ($D_0 > D_3 > D_1 > D_2$)

a. Fill the following truth table for 4 x2 Random Priority Encoder.

Inputs				Outputs		
D_0	D_3	D_1	D_2	A	B	Valid
0	0	0	0	X	X	0
1	X	X	X	0	0	1
0	1	X	X	1	1	1
0	0	1	X	0	1	1
0	0	0	1	1	0	1

b. Write the simplified expressions for outputs.

Handwritten Karnaugh map for output A:

$D_0 D_3$	$D_1 D_2$ 00	$D_1 D_2$ 01	$D_1 D_2$ 11	$D_1 D_2$ 10
00	X	1		
01	1	1	1	1
11				
10				

$$A = \overline{D_0} D_3 + \overline{D_0} \overline{D_1}$$

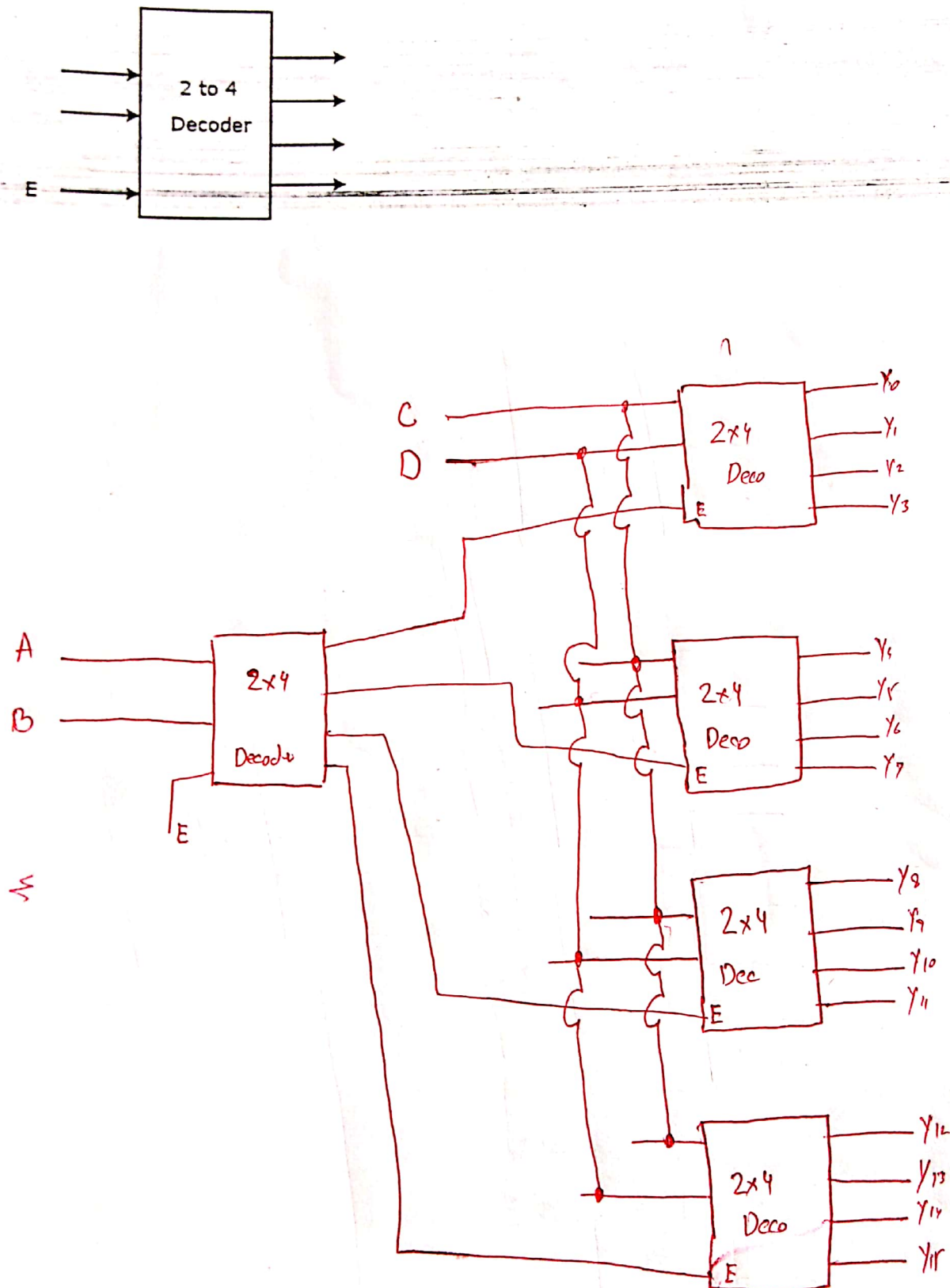
Handwritten Karnaugh map for output B:

$D_0 D_3$	$D_1 D_2$ 00	$D_1 D_2$ 01	$D_1 D_2$ 11	$D_1 D_2$ 10
00	X		1	1
01	1	1	1	1
11				
10				

$$B = \overline{D_0} D_3 + \overline{D_0} D_1$$

$$V = D_1 + D_2 + D_3 + D_4$$

Construct 4x16 line decoder using 2x4 line decoders. The block diagram of 2 x 4 line decoder is given below. The decoder shown below is implemented using 'AND' gates.



Implement given function using 4x1 multiplexer?

$$F(w, x, y, z) = \sum(1, 2, 5, 7, 8, 10, 11, 13, 15)$$

w	x	y	z	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

$$wx = 00$$

$$F = y \oplus z$$

$$wx = 01$$

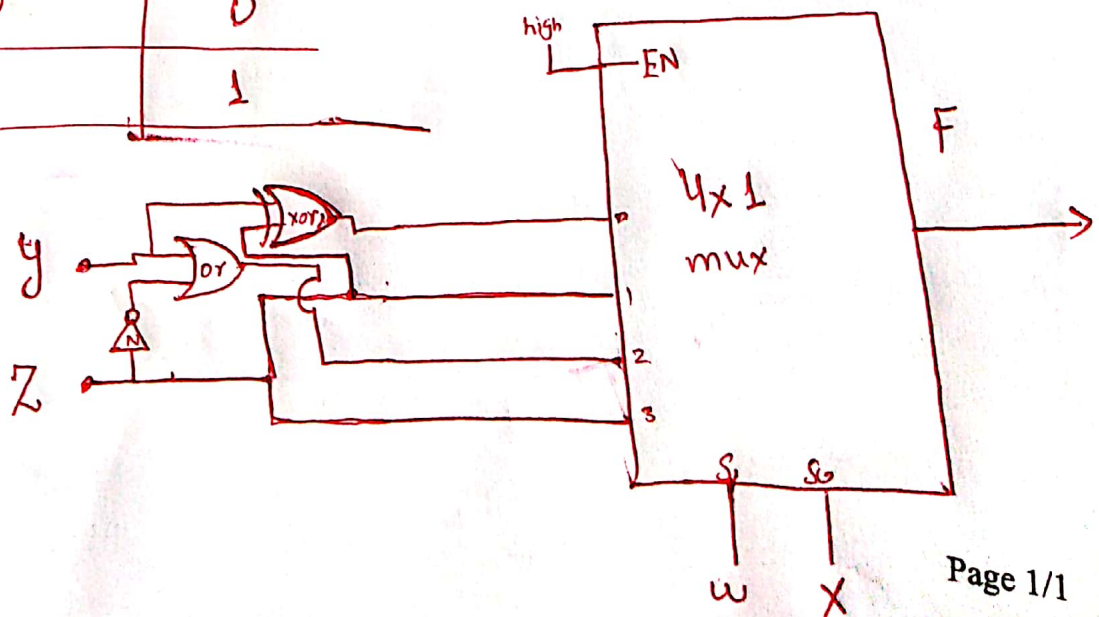
$$F = z$$

$$wx = 10$$

$$F = y + z'$$

$$wx = 11$$

$$F = z$$



Design 4x2 Random Priority Encoder with following input priority: ($D_0 > D_2 > D_1 > D_3$)

a. Fill the following truth table for 4.x2 Random Priority Encoder.

Inputs				Outputs		
D_0	D_2	D_1	D_3	A	B	Valid
0	0	0	0	X	X	0
1	X	X	X	0	0	1
0	1	X	X	1	0	1
0	0	1	X	0	1	1
0	0	0	1	1	1	1

b. Write the simplified expressions for outputs.

Handwritten Karnaugh Map for Output A:

	$D_1 D_3$	00	01	11	10
$D_0 D_2$	00	X	1		
	01	1	1	1	1
	11				
	10				

A

Handwritten Karnaugh Map for Output B:

	$D_1 D_3$	00	01	11	10
$D_0 D_2$	00	X	1	1	1
	01				
	11				
	10				

B

$$A = \overline{D_0} \overline{D_1} + \overline{D_0} D_2$$

$$B = \overline{D_0} \overline{D_2}$$

$$V = D_1 + D_2 + D_3 + D_4$$